

Linear Regression

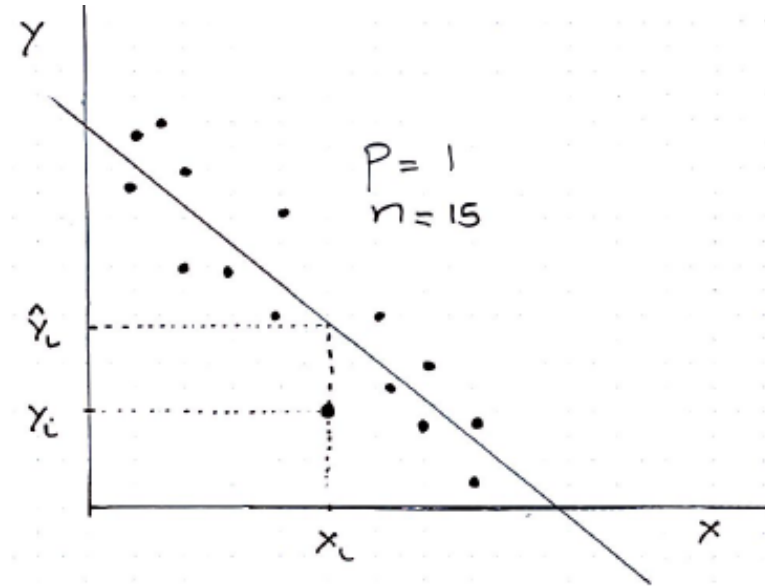


Figure 0.1: Linear model

Problem setup:

1. n = Number of observations
2. p = Number of predictor variables (excluding the constant 1)
3. y = Dependent variable in $\mathbb{R}$ (outcome)
4. $x_1, \dots, x_p$ = Independent variables (predictors)

We are interested in estimating the linear model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \epsilon,$$

where ϵ is the error term that models noise which is not captured by the predictor variables. The data consists of observations:

$$(y_i, x_{i1}, \dots, x_{ip}) \text{ for } i = 1, \dots, n.$$

The coefficients in the multiple linear regression model are chosen to minimize the sum of squared errors (residuals) given as:

$$\min_{\beta_0, \beta_1, \dots, \beta_p} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_p x_{ip})^2.$$

Key ideas:

1. Let us setup the optimization problem in vector and matrix notation as follows. Define:

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \mathbf{X} = \begin{bmatrix} 1 & x_{11} & \dots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{bmatrix}, \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}.$$

We can rewrite the linear model as:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

and formulate the optimization problem as:

$$\min_{\boldsymbol{\beta}} \underbrace{(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})}_{f(\boldsymbol{\beta})}.$$

This is a convex quadratic minimization problem where the global minimum can be found by setting the gradient to zero. Setting the gradient to zero we get:

$$\begin{aligned} \nabla_{\boldsymbol{\beta}} f(\boldsymbol{\beta}) &= \nabla_{\boldsymbol{\beta}} (\mathbf{y}'\mathbf{y} - \boldsymbol{\beta}'\mathbf{X}'\mathbf{y} - \mathbf{y}'\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}) \\ &= -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\boldsymbol{\beta} \\ &= \mathbf{0} \end{aligned}$$

Assuming $\mathbf{X}'\mathbf{X}$ is invertible, the optimal solution is given by:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

where the fitted values are $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$. We have the fitted values and the errors as:

$$\begin{aligned} \hat{y}_i &= \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \dots + \hat{\beta}_p x_{ip} \\ \epsilon_i &= y_i - \hat{y}_i. \end{aligned}$$

2. The optimal $\hat{\boldsymbol{\beta}}$ is an unbiased estimator of the true $\boldsymbol{\beta}$, namely $\mathbb{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$. The estimates have standard errors associated with them. This is based on the frequentist interpretation that we are developing the linear regression estimates using an observed data set that is sampled from a true population distribution. A common assumption is that the random observations y_i are independent of each other and have a constant variance denoted by σ^2 . Then we obtain:

$$\text{Variance}(\hat{\boldsymbol{\beta}}) = \text{Variance}((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\text{Variance}(\mathbf{y})(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = (\mathbf{X}'\mathbf{X})^{-1}\sigma^2$$

where $\text{Variance}(\hat{\boldsymbol{\beta}})$ is the variance-covariance matrix of the estimated coefficient vector $\hat{\boldsymbol{\beta}}$ and $\text{Variance}(\mathbf{A}\mathbf{y}) = \mathbf{A}\text{Variance}(\mathbf{y})\mathbf{A}'$ with $\text{Variance}(\mathbf{y}) = \sigma^2\mathbf{I}$ where $\mathbf{I}$ is the identity matrix. Here we think of $\mathbf{X}$ as a fixed matrix and $\mathbf{y}$ as random. Since the true variance σ^2 is unknown, we estimate it using the data from the regression fit as follows:

$$\hat{\sigma}^2 = \frac{1}{n-p-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

The division by the number $n-p-1$ is to ensure that the estimator is unbiased such that $\mathbb{E}(\hat{\sigma}^2) = \sigma^2$. The standard error of the coefficients is equal to the square root of the diagonal elements of the matrix $(\mathbf{X}'\mathbf{X})^{-1}\sigma^2$.

3. One of our main goals is to check whether a particular variable is useful in the model. We want to test the null hypothesis

$$H_0 : \beta_j = 0 \text{ versus } H_1 : \beta_j \neq 0.$$

The t-statistic is given as:

$$\text{t-statistic} = \frac{\hat{\beta}_j}{\text{Standard error}(\hat{\beta}_j)}.$$

If we estimate σ^2 by $\hat{\sigma}^2$, then:

$$\text{t-statistic} = \frac{\hat{\beta}_j}{\sqrt{((\mathbf{X}'\mathbf{X})^{-1})_{jj}\hat{\sigma}}}$$

The t-statistic follows a t-distribution with $n - p - 1$ degrees of freedom. If the absolute value of the t-statistic is high, the null hypothesis H_0 will be rejected in favor of H_1 . This indicates statistically that x_j is a significant predictor in the model and the p-value provides the probability of seeing a t-statistic as extreme as we observe under the null hypothesis.

Quality of fit

1. Let $\bar{y} = \sum_{i=1}^n y_i / n$. Define:

$$\text{SSE} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \text{ (Sum of squared errors or equivalently Sum of squared residuals)}$$

$$\text{SSR} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 \text{ (Sum of squares due to regression)}$$

$$\text{SST} = \sum_{i=1}^n (y_i - \bar{y})^2 \text{ (Total sum of squares)}$$

In linear regression, with the optimal estimates, we have:

$$\text{SST} = \text{SSE} + \text{SSR}.$$

The residual standard error is defined as $\sqrt{\text{SSE}/(n - p - 1)}$ and measures the lack of fit of the model. It is possible for models with more variables to have a higher residual standard error if the decrease in SSE is small relative to the increase in p (p increases implies SSE decreases and $n - p - 1$ decreases).

The proportion of the variance in the dependent variable that can be accounted by the variation in the independent variables is defined as R-squared or coefficient of determination:

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}} \text{ (R-squared or Coefficient of determination)}$$

R^2 is always between 0 and 1 and provides information on the goodness of the fit of the model. For example:

- (a) Regression fit is a horizontal line implies $R^2 = 0$ (the predictor variables have no explanatory power).
- (b) Regression fits perfectly all points on a straight line implies $R^2 = 1$ (the predictor variables have perfect explanatory power)
- (c) All the values of y_i lie in the same vertical line implies R^2 cannot be computed.

As we increase the number of predictor variables in the model, R^2 will never decrease (it will stay the same or increase). Hence it is important to be careful in using this to do model selection as you might overfit data. Furthermore, a good value of R^2 might be very different for a variety of applications. For example in finance, it is hard to predict stock prices and so even a useful model might have a small value of R^2 because the problem is challenging. On the other hand, a less useful model for an easier problem such as predicting revenue from the number of items sold might have a high R^2 .

For simple linear regression with a single variable:

$$y = \beta_0 + \beta_1 x + \epsilon,$$

the R^2 value is simply the $\text{Correlation}(x, y)^2$.

2. The adjusted R^2 statistic penalizes the R^2 statistic as more variables are added to the fit. The adjusted R^2 value can be negative and its value will always be lesser than or equal to R^2 . The adjusted R^2 increases when a new explanatory variable is added such that the increase in the fit is more than that expected by chance. The adjusted R^2 is one of the useful measures in selecting predictor variables during model selection. It is defined as:

$$\text{Adjusted } R^2 = 1 - \frac{\text{SSE}/(n - p - 1)}{\text{SST}/(n - 1)}$$

or equivalently

$$\text{Adjusted } R^2 = 1 - (1 - R^2) \left(\frac{n - 1}{n - p - 1} \right).$$

3. The F-statistic is used to test joint hypothesis. Let:

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0 \text{ versus}$$

$$H_1 : \text{At least one of the } \beta_j \text{ is nonzero}$$

The F-statistic is defined as:

$$\text{F-statistic} = \frac{(\text{SST} - \text{SSE})/p}{\text{SSE}/(n - p - 1)}$$

The F-statistic follows a F-distribution with p and $n - p - 1$ degrees of freedom.

Summary of output from linear regression in R

Function: `lm`

1. Residuals - This provides a summary of the residuals from the linear regression model. To access these for a model, use `model$residuals`.
2. Coefficients - This provides estimates of coefficients, standard error of coefficients, t-value and p-value ($P > |t|$). To access these use `model$coefficients` or `coefficients(model)`. For example, you can access the standard error by `coefficients(summary(model))["Std. Error"]`.
3. Residual standard error - This provides the average amount the response will deviate from the true regression line. It provides a measure of the lack of the fit of a linear model to the data. You can compute it as `sqrt(sum(model$residuals^2)/(n-p-1))`.
4. Multiple R-squared, Adjusted R-squared - R-squared is a measure between 0 and 1 to indicate the amount of variability explained by regression while adjusted R-squared accounts for number of predictors.
5. F-statistic and p-value - Test to see if at least one of the predictors is nonzero.

Additional Notes

1. Why is $\hat{\beta}$ an unbiased estimator of the true β in $\mathbf{y} = \mathbf{X}\beta + \epsilon$?

The optimal solution is given by:

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

Then:

$$\begin{aligned}\mathbb{E}[\hat{\beta}] &= \mathbb{E}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}] \\ &= \mathbb{E}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \epsilon)] \\ &= \mathbb{E}[\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\epsilon] \\ &= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbb{E}[\epsilon] \\ &\quad [\text{since } \beta \text{ and } \mathbf{X} \text{ are fixed}] \\ &= \beta \\ &\quad [\text{since } \mathbb{E}[\epsilon] = \mathbf{0}].\end{aligned}$$

2. Why is $\hat{\sigma}^2$ an unbiased estimator of σ^2 , namely $\mathbb{E}[\hat{\sigma}^2] = \sigma^2$?

The formula for $\hat{\sigma}^2$ is given by:

$$\hat{\sigma}^2 = \frac{1}{n-p-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

The optimal solution is given by:

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

Then:

$$\begin{aligned} \mathbb{E} \left[\sum_{i=1}^n (y_i - \hat{y}_i)^2 \right] &= \mathbb{E} \left[(\mathbf{y} - \mathbf{X}\hat{\beta})'(\mathbf{y} - \mathbf{X}\hat{\beta}) \right] \\ &= \mathbb{E} \left[\mathbf{y}'\mathbf{y} - \hat{\beta}'\mathbf{X}'\mathbf{y} - \mathbf{y}'\mathbf{X}\hat{\beta} + \hat{\beta}'\mathbf{X}'\mathbf{X}\hat{\beta} \right] \\ &= \mathbb{E} \left[\mathbf{y}'\mathbf{y} - \mathbf{y}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} - \mathbf{y}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} + \mathbf{y}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \right] \\ &= \mathbb{E} \left[\mathbf{y}'\mathbf{y} - \mathbf{y}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \right] \\ &= \mathbb{E} \left[\mathbf{y}'(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')\mathbf{y} \right] \\ &= \mathbb{E} \left[(\mathbf{X}\beta + \epsilon)'(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')(\mathbf{X}\beta + \epsilon) \right] \\ &\quad [\text{since } \mathbf{y} = \mathbf{X}\beta + \epsilon] \\ &= \mathbb{E} \left[\epsilon'(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')\epsilon \right] + \beta'\mathbf{X}'(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')\mathbf{X}\beta \\ &\quad [\text{since } \mathbb{E}[\epsilon] = \mathbf{0}] \\ &= \mathbb{E} \left[\epsilon'(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')\epsilon \right] + \underbrace{\beta'\mathbf{X}'\mathbf{X}\beta - \beta'\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta}_{=0} \\ &= \left[\text{trace}(\mathbf{I}_n) - \text{trace}(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') \right] \sigma^2 \\ &\quad [\text{since } \mathbb{E}[\epsilon_i^2] = \sigma^2 \text{ for all } i \text{ and } \mathbb{E}[\epsilon_i\epsilon_j] = 0 \text{ for } i \neq j \text{ where } \mathbf{I}_n \text{ is a } n \times n \text{ identity matrix}] \\ &= \left[\text{trace}(\mathbf{I}_n) - \text{trace}(\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}) \right] \sigma^2 \\ &\quad [\text{since } \text{trace}(\mathbf{AB}) = \text{trace}(\mathbf{BA}) \text{ when } \mathbf{A} \text{ and } \mathbf{B} \text{ are } m \times n \text{ and } n \times m \text{ matrices}] \\ &= \left[\text{trace}(\mathbf{I}_n) - \text{trace}(\mathbf{I}_{p+1}) \right] \sigma^2 \\ &\quad [\text{since } \mathbf{X} \text{ is a } n \times (p+1) \text{ matrix where } \mathbf{I}_{p+1} \text{ is a } (p+1) \times (p+1) \text{ identity matrix}] \\ &= (n-p-1)\sigma^2. \end{aligned}$$

3. SST = SSR + SSE. Why?

We solve:

$$\min_{\beta_0, \beta_1, \dots, \beta_p} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_p x_{ip})^2.$$

Setting derivative with respect to β_0 to 0 implies the optimal estimates satisfy:

$$\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_p x_{ip}) = 0$$

which is equivalent to

$$\sum_{i=1}^n (y_i - \hat{y}_i) = \sum_{i=1}^n \epsilon_i = 0.$$

Similarly taking derivative with respect to β_j for each $j = 1, \dots, p$, the optimal estimates satisfy:

$$\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_p x_{ip}) x_{ij} = 0$$

which is equivalent to

$$\sum_{i=1}^n (y_i - \hat{y}_i) x_{ij} = \sum_{i=1}^n \epsilon_i x_{ij} = 0.$$

Now we observe that:

$$\begin{aligned} \text{SST} &= \sum_{i=1}^n (y_i - \bar{y})^2 \\ &= \sum_{i=1}^n (y_i - \hat{y}_i + \hat{y}_i - \bar{y})^2 \\ &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + 2 \sum_{i=1}^n (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) \\ &= \text{SSE} + \text{SSR} + 2 \sum_{i=1}^n \epsilon_i (\hat{y}_i - \bar{y}) \\ &= \text{SSE} + \text{SSR} + 2 \sum_{i=1}^n \epsilon_i \hat{y}_i - 2 \sum_{i=1}^n \epsilon_i \bar{y} \\ &= \text{SSE} + \text{SSR} + 2 \sum_{i=1}^n \epsilon_i (\hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \dots + \hat{\beta}_p x_{ip}) - 2 \sum_{i=1}^n \epsilon_i \bar{y} \\ &= \text{SSE} + \text{SSR} + 2 \sum_{i=1}^n \epsilon_i \hat{\beta}_0 + 2 \sum_{i=1}^n \epsilon_i \hat{\beta}_1 x_{i1} + \dots + 2 \sum_{i=1}^n \epsilon_i \hat{\beta}_p x_{ip} - 2 \sum_{i=1}^n \epsilon_i \bar{y} \\ &= \text{SSE} + \text{SSR} + 0 + 0 + \dots + 0 - 0 \\ &= \text{SSE} + \text{SSR}. \end{aligned}$$

4. $R^2 = \text{Correlation}(x, y)^2$ in simple linear regression. Why?

We solve:

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2.$$

Setting derivative with respect to β_0 to 0 implies the optimal estimates satisfy:

$$\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0 \text{ which is equivalent to } \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where $\bar{x} = \sum_{i=1}^n x_i / n$ and $\bar{y} = \sum_{i=1}^n y_i / n$. Setting derivative with respect to β_1 to 0, we get:

$$\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) x_i = 0$$

This gives:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n y_i x_i - n \bar{y} \bar{x}}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}.$$

Now we observe that:

$$\begin{aligned} R^2 &= \frac{\text{SSR}}{\text{SST}} \\ &= \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \\ &= \frac{\sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 x_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \\ &= \frac{\sum_{i=1}^n (\bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \\ &= \hat{\beta}_1^2 \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \\ &= \left(\frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \right)^2 \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \\ &= \frac{(\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}))^2}{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2} \\ &= \text{Correlation}(x, y)^2. \end{aligned}$$

5. Maximum likelihood estimation version of linear regression:

We consider an alternative formulation where:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon_i,$$

where ϵ_i for $i = 1, \dots, n$ consists of independent and identically distributed normal random variables with mean 0 and variance σ^2 . The likelihood of seeing the data $(y_i, x_{i1}, \dots, x_{ip})$ for $i = 1, \dots, n$ is given by:

$$\prod_{i=1}^n \underbrace{f(y = y_i | x_{i1}, \dots, x_{ip}, \beta, \sigma^2)}_{\text{normal probability density function}} = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - (\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip}))^2}{2\sigma^2}}$$

We want to find β and σ^2 to maximize the likelihood or equivalently to maximize the log-likelihood where the log-likelihood is defined as:

$$\text{Log-Likelihood}(\beta, \sigma^2) = LL(\beta, \sigma^2) = -\sum_{i=1}^n \frac{(y_i - (\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip}))^2}{2\sigma^2} - \frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2).$$

Find the optimal β is equivalent here to solving the minimization problem (since the choice of σ^2 does not affect the solution of β):

$$\min_{\beta} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip}))^2$$

where the optimal solution is given by

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

Plugging this back in we see finding the optimal σ^2 then reduces to solving:

$$\max_{\sigma^2} -\sum_{i=1}^n \frac{(y_i - \hat{y}_i)^2}{2\sigma^2} - \frac{n}{2} \log(\sigma^2)$$

Setting derivative to zero we get:

$$\sum_{i=1}^n \frac{(y_i - \hat{y}_i)^2}{2\sigma^4} - \frac{n}{2\sigma^2} = 0$$

which gives:

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

This is exactly the average of the squared deviations between the true value and the predicted values.